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The cognitive impact of ChatGPT in higher education: A systematic review of critical and creative thinking outcomes

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ABSTRACT

This systematic review examines how ChatGPT influences university students' critical and creative thinking by synthesizing 67 empirical studies (2022-2025) following PRISMA guidelines. We applied a dual-lens framework distinguishing convergent (critical thinking) and divergent (creative thinking) processes. Analysis revealed that, ChatGPT supports cognitive development when embedded within inquiry-oriented and scaffolded instructional designs, especially through processes such as metacognitive regulation, argumentative reasoning, and idea generation, which in turn produce synergistic gains across both critical and creative domains. However, in unstructured contexts, researchers observed asymmetrical patterns favoring creativity over critical thinking and joint declines across domains, both of which often triggered cognitive offloading. These findings highlight that ChatGPT's cognitive effects are contingent on pedagogical framing. We recommend embedding explicit scaffolds—such as reflective prompts, rubric-guided evaluation, and AI-literacy training—to harness ChatGPT's potential as a dialogic partner rather than a convenience tool. Implications for instructional design involve co-activating critical and creative processes through recursive inquiry tasks and multi-source feedback loops. This review informs educators and researchers on designing AI-mediated learning environments that promote higher-order thinking while mitigating risks of dependency and reduced epistemic engagement.

1. Introduction

ChatGPT has rapidly reshaped higher education since its 2022 launch, reaching over 100 million users within two months and becoming widely embedded in academic practices such as writing support, brainstorming, formative feedback, and inquiry-based learning (Deng et al., 2024; Dimeli & Kostas, 2025; Hu, 2023; Tlili et al., 2023). This widespread adoption has prompted critical discourse about cognitive and pedagogical implications, particularly concerning students' higher-order thinking capacities (Lo et al., 2024; Vargas-Murillo et al., 2023; Zirar, 2023).

Research on ChatGPT's educational impact reveals contradictory findings. Some studies report improvements in reasoning, argumentation, and idea generation (Daniel et al., 2025; Dimeli & Kostas, 2025; Farazouli et al., 2023), while others highlight concerns including diminished creativity, superficial analytical depth, and cognitive offloading from excessive AI reliance (Bai et al., 2023; Roy & Putatunda, 2023; Vargas-Murillo et al., 2023). These divergent findings suggest that the influence of ChatGPT on student cognition is not intrinsic to the tool itself but contingent upon pedagogical framing, task design, and instructional scaffolding (Deng et al., 2024; García-López et al., 2025; Labadze et al., 2023).

Existing systematic reviews have examined ChatGPT's educational role through varied thematic lenses (Table 1). Zirar (2023) explored its use in learning and assessment, noting both problem-solving potential and risks of student overreliance. Deng et al. (2024) conducted a meta-analysis of experimental studies, highlighting improvements in academic performance, motivation, and higher-order thinking, though without disaggregating specific cognitive processes. Lo et al. (2024) reviewed behavioral, emotional, and cognitive engagement, reporting mixed findings, particularly regarding critical thinking. Melisa et al. (2025) focused on critical thinking, emphasizing evaluation and independent judgment but without broader cognitive integration. Dimeli and Kostas (2025) discussed gains in "cognitive performance," yet did not operationalize specific constructs. Other reviews, such as García-López et al. (2024), Labadze et al. (2023), and Vargas-Murillo et al.

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Table 1
Existing systematic reviews on ChatGPT and cognitive outcomes.

Authors (Year)	Review Focus	Cognitive Aspects	Study Count
Dimeli and Kostas (2025)	ChatGPT in education	General cognitive performance	50 studies
Deng et al. (2024)	Academic outcomes	Higher-order thinking	Meta-analysis
Lo et al. (2024)	Student engagement	Cognitive engagement	72 studies
Zirar (2023)	Learning/ assessment	Idea generation, Problem-solving, reflection	25 studies
García-López et al. (2025)	AI-assisted learning	Critical thinking concerns	20 studies
Farazouli et al. (2023)	Academic skills development	Reasoning, creativity	18 studies

(2023), addressed ethical, pedagogical, and infrastructural dimensions, with limited attention to cognitive specificity. Collectively, these reviews provide valuable foundations but tend to emphasize broad learning outcomes or general cognitive domains.

Critically, no existing review has systematically examined how distinct forms of higher-order thinking are differentially affected. In particular, the distinction between convergent cognitive processes (e.g., critical thinking) and divergent cognitive processes (e.g., creative thinking) provides a theoretically grounded lens for understanding how ChatGPT may independently shape these domains and influence their interaction within AI-mediated learning environments.

This review addresses the gap by introducing a dual-lens framework of critical and creative thinking (CCT) as complementary yet distinct domains of higher-order cognition. Critical thinking (CT) involves interpreting and evaluating information, making reasoned judgments, and constructing evidence-based arguments (Ennis, 2018; Facione, 2011). Creative thinking (CrT) encompasses producing original, flexible, and valuable ideas through divergent thinking, synthesis, and idea generation (Runco & Acar, 2012; Torrance, 1974). These competencies are interdependent in academic contexts requiring both analytical rigor and innovative problem-solving (Brookhart, 2010; Cropley, 2006). This integrated perspective is particularly relevant in AI-mediated environments, reflecting growing recognition of these competencies as foundational to academic inquiry and adaptive expertise. By applying this dual-lens framework, the review systematically maps cognitive outcomes while attending to instructional design, task structure, and contextual moderators.

This study systematically analyzes peer-reviewed empirical research on ChatGPT's influence on students' CCT in higher education settings. The review aims to: (1) clarify the nature and extent of ChatGPT's cognitive influence; (2) identify pedagogical conditions supporting or constraining higher-order thinking; and (3) generate actionable insights for instructional practice and future research. Accordingly, the study is guided by the following research questions:

RQ1: What are the characteristics of empirical studies examining ChatGPT's influence on students' critical and creative thinking?
RQ2: How does ChatGPT influence students' critical thinking under different instructional conditions?
RQ3: How does ChatGPT influence students' creative thinking under different instructional conditions?
RQ4: What co-occurrence patterns emerge when both thinking domains are examined together, and how do instructional factors moderate these relationships?

2. Conceptual framework

This section outlines the theoretical foundations that inform our dual-lens framework for understanding ChatGPT's influence on critical and creative thinking in higher education. We first establish the complementary relationship between these cognitive domains, then present an integrated set of perspectives that explain how ChatGPT can mediate cognition in educational contexts.

2.1. Critical and creative thinking as complementary cognitive domains

Critical and creative thinking represent distinct yet interdependent cognitive processes essential for academic success and adaptive expertise. Critical thinking involves interpretive, analytical, and evaluative judgement, enabling students to assess arguments, weigh evidence, and construct reasoned responses (Ennis, 2018; Facione, 2011). Creative thinking encompasses generative cognition—producing original and flexible ideas through fluency, synthesis, and elaboration (Runco & Acar, 2012; Torrance, 1974).

While conceptually distinct, these processes function synergistically in academic inquiry: creative thinking generates possibilities, while critical thinking filters and refines them through structured analysis (Brookhart, 2010; Cropley, 2006). This interplay becomes particularly significant in AI-enhanced learning environments, where tools like ChatGPT can both stimulate ideation and risk promoting cognitive shortcuts (Bai et al., 2023; Daniel et al., 2025). Understanding this dynamic relationship provides the foundation for examining how ChatGPT differentially affects convergent (critical) and divergent (creative) thinking processes.

Within the broader field of educational technology, CT and CrT are increasingly recognized as foundational digital-age competencies (OECD, 2019, pp. 23–35; Trilling, 2009) that shape how learners interact with intelligent systems. CT aligns with the epistemic and evaluative dimensions of human–AI interaction—such as verifying AI-generated content, critiquing reasoning structures, and maintaining academic integrity. CrT, on the other hand, reflects the generative affordances of AI—using tools like ChatGPT for brainstorming, rephrasing, and ideational recombination. Scholars such as Kirschner and van Merriënboer (2013) emphasize the importance of scaffolding these cognitive processes in technology-mediated contexts to avoid surface learning and passive consumption. In this review, CT and CrT are treated not simply as psychological traits, but as situated cognitive performances shaped by AI affordances, instructional design, and pedagogical framing (Holmes et al., 2022). This reconceptualization is critical for understanding how generative tools like ChatGPT mediate higher-order thinking in contemporary higher education settings.

2.2. Theoretical foundations for AI-mediated cognition

To interpret the diverse cognitive effects of ChatGPT in higher education, this review draws on an integrated framework spanning cognitive, sociocultural, and technological dimensions. These complementary perspectives provide interpretive lenses for understanding how ChatGPT mediates students' critical and creative thinking within varied instructional environments.

AI Literacy serves as a foundational moderator, influencing whether students engage with ChatGPT reflectively or superficially. This competency—defined as the ability to interpret, evaluate, and responsibly use AI systems (Long & Magerko, 2020; Ng et al., 2021)—shapes how students craft prompts, interpret outputs, and regulate dependence on AI assistance. In this review, AI Literacy informs our analysis of how prompt quality, epistemic vigilance, and verification practices moderate both critical and creative thinking.

Cognitive Regulation Theories explain mechanisms through which ChatGPT supports or constrains higher-order thinking. Self-Regulated Learning (SRL) (Zimmerman, 2002) refers to learners' capacity to plan, monitor, and regulate their learning through strategic goal-setting, reflection, and adaptation. Metacognitive frameworks emphasize awareness of one's thinking processes and the ability to evaluate and adjust cognitive strategies (Azevedo et al., 2010). Both perspectives are crucial for understanding how students interact with ChatGPT as a thinking aid—whether they critically interrogate its outputs or passively adopt them. Cognitive Load Theory (Sweller, 2011) complements this view by explaining how AI tools may reduce extraneous effort, freeing cognitive resources for analysis or synthesis—but only when embedded in well-structured tasks. Together, these theories clarify when ChatGPT promotes productive cognitive engagement and when it enables shallow processing or offloading.

External Cognition Frameworks conceptualise thinking as extending beyond individual minds to include tools and networks. Drawing on Distributed Cognition (Hutchins, 1995), ChatGPT can be interpreted as an interactive cognitive artifact that can support reasoning when learners engage in iterative cycles of generation, critique, and refinement. In such cycles, outputs function as provisional representations that students evaluate, restructure, and justify—activities that can support both convergent judgement (critical thinking) and divergent exploration (creative thinking). From a Connectivist lens (Siemens, 2005), ChatGPT can be analyzed as a resource within distributed learning networks that supports pattern detection and cross-context recombination, especially in tasks that demand synthesis across contexts rather than surface-level fluency.

Activity and Context Theories address why cognitive outcomes vary across implementations. Cultural Historical Activity Theory (CHAT) (Engeström, 1987) conceptualises learning as an activity system oriented toward an object (task goal) and mediated by tools, rules, community, and division of labour. Applied here, CHAT provides a way to analyse when and how ChatGPT reshapes learning activity—for example by shifting the division of labour (what students do vs. what the tool produces), altering rules around verification and attribution, and changing the object of tasks from producing an answer to producing a justified, iteratively refined argument. Boundary Object Theory (Star & Griesemer, 1989) captures ChatGPT's interpretive flexibility—maintaining coherence while enabling contextual adaptation across disciplines and tasks—helping explain how the same tool can support standardised academic functions and personalized creative expression.

Conceptually, these perspectives operate as an explanatory constellation: AI Literacy shapes engagement quality; Cognitive Regulation Theories clarify proximal mechanisms (monitoring, load management, and offloading); External Cognition Frameworks explain how reasoning can be distributed across learner–tool dialogue and broader knowledge networks; and Activity and Context Theories explain why outcomes differ across pedagogical ecosystems and institutional rules. For example, when a student uses ChatGPT to draft an essay (External Cognition), their ability to critically evaluate the output depends on AI Literacy, while the quality of their revision reflects Metacognitive Regulation. Whether this process enhances or undermines critical thinking is shaped by the Activity System—including task rules (e.g., must cite sources), division of labor (who generates vs. evaluates content), and instructional scaffolds (e.g., rubrics, reflection prompts).

Collectively, these perspectives treat ChatGPT neither as an inherently beneficial “thinking agent” nor as a neutral instrument, but as an interactive cognitive artifact within a socio-technical learning system. Its cognitive influence emerges from how learners and instructors mobilise its affordances through prompting, critique, verification, and revision, and from the surrounding pedagogical rules, feedback structures, and learning goals. Taken together, the framework supports interpretation of the diverse outcomes across the 67 studies, including patterns of synergistic enhancement, asymmetrical development, and joint erosion of critical and creative thinking capabilities.

3. Method

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021) to examine ChatGPT's influence on students' critical and creative thinking in higher education.

3.1. Information sources and search strategy

To systematically identify empirical studies examining the influence of ChatGPT on students' CCT in higher education, a comprehensive search strategy was developed. Key conceptual terms were identified across three clusters—ChatGPT technologies, CCT constructs, and higher education contexts—and combined using Boolean operators, as presented in Table 2. Synonyms and related terms were systematically included to account for diverse terminologies employed across the literature and maximize the retrieval of relevant studies.

The finalized search string was applied to three major academic databases: Web of Science (SSCI collection), Scopus, and ERIC, selected for their extensive coverage of peer-reviewed research in education, cognitive science, and interdisciplinary fields. The database search was conducted on April 19, 2025. All retrieved records were exported into a reference management tool for duplicate removal and initial eligibility screening. The search strategy was designed to balance sensitivity and specificity, ensuring comprehensive coverage of studies directly investigating ChatGPT's cognitive influence on university and college students.

While the search strategy was designed to ensure comprehensive coverage, the inclusion of only peer-reviewed journal articles and English-language publications may introduce publication and language bias. Studies reporting null or negative findings, or conducted in non-English contexts, may be underrepresented in the final dataset. These constraints are further addressed in the limitations.

3.2. Inclusion and exclusion criteria

Guided by the research questions, predefined inclusion and exclusion criteria were established to ensure the relevance, methodological rigor, and conceptual alignment of the selected studies. Only empirical studies published between January 1, 2022 and April 19, 2025 were considered, capturing research conducted after the release and rapid adoption of ChatGPT technologies in educational contexts. Following established systematic review practices, only peer-reviewed journal articles were included to ensure a minimum threshold of methodological rigor and quality assurance, while other sources such as preprints, conference proceedings, and dissertations were excluded (Korpershoek et al., 2016).

Articles were eligible for inclusion if they explicitly focused on the use of ChatGPT (or related models such as GPT-3 or GPT-4) and investigated its influence on students' CT, CrT, higher-order thinking, or related cognitive skills within higher education settings. Studies needed to report original empirical data, either quantitative, qualitative, or mixed-methods, collected from university or college students at the undergraduate or postgraduate level, or from educators and stakeholders reporting on ChatGPT's influence on students' CCT. Conceptual articles, opinion pieces, theoretical essays, and reviews without primary data collection were excluded to maintain the focus on empirically grounded evidence (Booth et al., 2016). Additionally, only English-language publications were included to avoid potential issues

Table 2
Search strings and boolean operators.

Terms	Search strings
ChatGPT	(“ChatGPT” OR “GPT-4″ OR “GPT 4″ OR “GPT-3″ OR “GPT 3″)
Critical and Creative thinking	(“Critical thinking” OR “Creative thinking” OR “Higher-order thinking” OR “Cognitive skills” OR “Problem-solving” OR “Creative problem-solving” OR “Divergent thinking” OR “Analytical thinking” OR “Innovation"
Higher Education	“Higher education” OR “University students” OR “College students” OR “Postsecondary education” OR “Tertiary education"

related to translation bias (Moher et al., 2009). Table 3 summarizes the inclusion and exclusion criteria applied when selecting the articles.

3.3. Screening and selection process

Fig. 1 presents the PRISMA flow diagram illustrating the selection process. The initial database search retrieved 409 records: Web of Science (n = 161), Scopus (n = 177), and ERIC (n = 71). After removing 185 duplicates, 224 records were screened by title and abstract, resulting in 96 full-text articles for eligibility review.

Two reviewers independently screened all full texts based on predefined inclusion and exclusion criteria. Inter-rater agreement was initially 95%; discrepancies were resolved through discussion to reach full consensus (Hallgren, 2012).

Twenty-seven studies were excluded during full-text review: 6 were non-empirical, 4 addressed non-higher education settings, 9 did not assess ChatGPT's influence on learning outcomes, and 8 lacked empirical examination of CT or CrT. An additional two studies were excluded after quality appraisal due to methodological limitations.

A total of 67 empirical studies met all criteria and were included in the final synthesis. This rigorous process ensured the review integrated high-quality, relevant evidence on ChatGPT's cognitive impact in higher education.

3.4. Quality appraisal

The methodological quality of included studies was assessed using the Mixed Methods Appraisal Tool (MMAT, 2018; Hong et al., 2018), which evaluates qualitative, quantitative, and mixed-methods research across five criteria: (1) alignment of design and research question, (2) sampling adequacy, (3) data collection appropriateness, (4) outcome data completeness, and (5) analytical rigor. This tool is widely used for appraising methodological quality in interdisciplinary reviews (Pluye et al., 2011).

Two reviewers independently applied the MMAT, rating each study with "Yes," "No," or "Can't tell" across the five criteria. Inter-rater reliability was assessed to evaluate the consistency of independent quality ratings. The raw agreement rate across all criteria was 89%, and Cohen's kappa coefficient was $\kappa = 0.77$, indicating substantial agreement (Landis & Koch, 1977). Discrepancies were resolved through discussion until consensus was reached. Only studies meeting at least three of the five MMAT criteria were retained for synthesis.

Two studies were excluded due to methodological limitations. Among the 67 retained, 41 (61%) met all five criteria, 21 (31%) met four, and 5 (7%) met three. Common limitations involved incomplete outcome reporting and insufficient analytic detail. The MMAT process ensured inclusion of studies with acceptable methodological rigor, enhancing the credibility of the synthesis.

Table 3
Inclusion criteria for article selection.

Category	Inclusion Criteria	Rationale
Focus	Studies focusing on ChatGPT (or GPT-3, GPT-4)	To ensure the review isolates the cognitive influence of ChatGPT technologies.
Constructs	Examines students' critical, creative, or higher-order thinking	To align with the review's focus on cognitive skills development.
Population	University or college students	To specifically investigate ChatGPT's impact on higher education learners.
Context	Higher education settings	To ensure contextual consistency in learning environments.
Study Type	Empirical research studies	To focus on studies providing original empirical evidence.
Language	English publications	To ensure consistency in language comprehension and synthesis.
Publication Date	2022–2025	To capture the contemporary research landscape after ChatGPT's release.
Document Type	Peer-reviewed journal articles	To ensure scholarly rigor and publication quality.

3.5. Data extraction and coding

The data extraction and coding process was designed to ensure methodological transparency and alignment with the study's research questions and conceptual framework. A primarily deductive approach guided the identification of analytical categories, informed by the four predefined research questions (Booth et al., 2016; Thomas & Harden, 2008). Limited inductive flexibility allowed for emergent insights (Sandelowski, 2010). A codebook specifying operational definitions and classification criteria was iteratively developed and is provided in the Supplementary Appendix.

For RQ1, extracted data included study design, discipline, region, year, participants, instructional application of ChatGPT, and CCT assessment strategies. Additional variables captured ChatGPT model versions (e.g., GPT-3.5 or GPT-4), instructional modalities (e.g., guided vs. exploratory use), and operational definitions of CCT. While not systematically reported in the results due to heterogeneity, these features informed comparative interpretations discussed in the limitations.

For RQ2, coding addressed impact direction (positive/negative/mixed), evidence types (e.g., standardized tests, student artifacts, self-reports), specific CT effects (e.g., reasoning, argumentation), instructional context, ChatGPT role, and reported moderators (e.g., scaffolding, reflection).

For RQ3, extracted elements included direction and type of impact, evidence credibility (e.g., creative products, perception data), and changes in fluency, originality, elaboration, or flexibility. Coding also covered pedagogical strategies (e.g., prompt design, reflective tasks), ChatGPT functions (e.g., ideation, co-creation), and contextual moderators (e.g., digital literacy, affective risk).

For RQ4 (co-occurrence of CT and CrT), data included the nature of interaction (synergistic, asymmetrical, suppressive), evidence types, analytical strategies, and pedagogical conditions (e.g., epistemic prompts, scaffolds, framing). Moderators and author-noted implications regarding CT-CrT interplay were also recorded.

A coding matrix was developed in Excel, with rows representing studies and columns denoting extracted variables. The process proceeded in three phases: (1) a pilot extraction on 10% of studies by both reviewers; (2) full extraction by the first author with 30% double-coded and verified by the second reviewer; and (3) a final cross-check for consistency. Discrepancies were resolved by discussion. Triangulation and calibration throughout ensured rigor, and synthesized data were organized into descriptive summaries and thematic categories to support narrative synthesis.

3.6. Ethical considerations

This systematic review analyzed published research and required no ethical approval. All included studies were peer-reviewed publications available in the public domain.

4. Results

4.1. Characteristics of included studies

This systematic review synthesized 67 empirical studies examining ChatGPT's influence on students' CCT in higher education. Following rigorous screening and quality appraisal, the included studies reflected diverse methodological designs, disciplinary domains, geographic regions, participant groups, instructional application of ChatGPT, and assessment approaches (Table 4). Among them, 23 studies focused solely on CT, 11 on CrT, and 33 addressed both constructs concurrently.

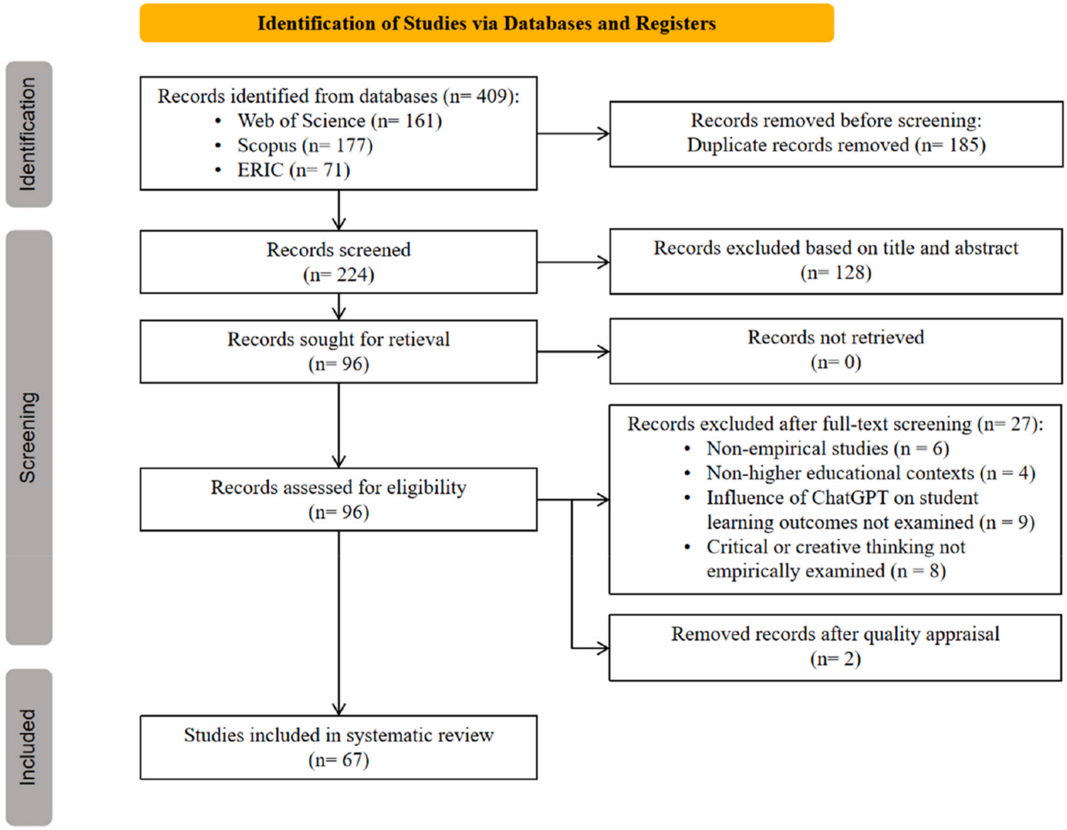

Fig. 1. PRISMA flow diagram showing the systematic selection of studies examining ChatGPT's influence on critical and creative thinking in higher education (adapted from Page et al., 2021).

4.1.1. *Study design types*

The studies employed varied methodological approaches, reflecting the evolving inquiry into AI integration in education. Specifically, 26 studies (39%) utilized quantitative designs, including surveys, experiments, and pre-post assessments of CCT outcomes. Qualitative methods were employed in 20 studies (30%), featuring interviews, thematic analyses, and content analysis of student artifacts. A further 21 studies (31%) adopted mixed-methods designs, combining quantitative assessments (e.g., rubrics, scales) with qualitative insights (e.g., reflections, observations) to capture both measurable outcomes and experiential dimensions of student thinking.

4.1.2. *Disciplinary focus*

The studies spanned a broad range of academic disciplines, highlighting the interdisciplinary nature of ChatGPT research in higher education. STEM fields accounted for the largest proportion of studies (n = 24, 36%), followed by general education and teacher education (n = 21, 31%) and language and writing disciplines (n = 18, 27%). An additional 4 studies (6%) were classified as interdisciplinary or from other fields, such as project-based learning or cross-institutional initiatives. This distribution underscores the cross-cutting interest in ChatGPT as a pedagogical tool across both technical and humanistic domains.

4.1.3. *Regional distribution*

The geographical distribution of studies demonstrated the global reach of ChatGPT's educational applications. Regional classification was based on the geographical location of the participant sample or the institutional setting where the study was conducted. The highest concentration was in Asia (n = 39, 58%), followed by Europe (n = 12, 18%), North America (n = 7, 10%), Oceania (n = 3, 4%), and Africa (n = 3, 4%). Several studies involved cross-national or multinational collaborations and were counted in multiple regions. The prominence of Asian studies reflects the rapid adoption and experimentation with AI tools across higher education systems in the region.

4.1.4. *Publication trends*

The publication trend aligns with the surge of academic interest following the public release of ChatGPT. Of the 67 included studies, 7 (10%) were published in 2023, 39 (58%) in 2024, and 21 (31%) in 2025 (up to April). The most frequent publication venues included Education and Information Technologies (n = 4), Frontiers in Education (n = 3), Journal of Information Technology Education: Research (n = 3), and Computers & Education (n = 3), indicating a strong orientation toward technology-enhanced learning and interdisciplinary educational innovation.

4.1.5. *Participant characteristics*

Participants across the studies reflected a broad spectrum of higher education stakeholders. Undergraduate students were the most common participant group (n = 45, 67%), reflecting widespread integration of ChatGPT into university-level coursework, assignments, and assessments. Graduate students featured in 43 studies (64%), particularly within specialized, research-oriented, or professional programs such as teacher education, engineering, and applied linguistics. Additionally, 11 studies (16%) included educators or faculty participants, often in the context of instructional innovation, AI-facilitated pedagogy, or professional development. This distribution highlights the expanding interest in ChatGPT's pedagogical potential across diverse learner populations and educational roles.

4.1.6. *Instructional applications of ChatGPT*

The studies revealed diverse instructional applications of ChatGPT, reflecting emerging pedagogical patterns in AI-integrated learning environments. The most common use was for brainstorming and idea

Table 4
Descriptive summary of included studies in the review (n = 67).

	Frequency	Percent
Study Focus		
Critical Thinking (CT) only	23	34.33
Creative Thinking (CrT) only	11	16.42
Both CT and CrT	33	49.25
Study Design Type		
Quantitative	26	38.81
Qualitative	20	29.85
Mixed Methods	21	31.34
Disciplinary Focus		
STEM	24	35.82
General Education/Teacher Education	21	31.34
Language and Writing	18	26.87
Interdisciplinary/Other	4	5.97
Geographic Region		
Asian	39	58.21
Europe	12	17.91
North America	7	10.45
Oceania	3	4.48
Africa	3	4.48
Year of publication		
2023	7	10.45
2024	39	58.21
2025 (up to April)	21	31.34
Common Journals		
Education and Information Technologies	4	5.97
Frontiers in Education	3	4.48
Journal of Information Technology Education: Research	3	4.48
Computers & Education	3	4.48
Participants		
Undergraduate students	45	67.16
Graduate students	43	64.18
Educators/Faculty	11	16.42
Instructional Applications of ChatGPT		
Brainstorming and Idea Generation	20	29.85
Feedback and Reflection	19	28.36
Writing Support	12	17.91
Flipped or Blended Classrooms	8	11.94
Problem-Solving/Computation	5	7.46
Critical and Creative Thinking Assessment Strategy		
Direct Assessment (e.g., TTCT, performance tasks)	27	40.30
Indirect Assessment (e.g., perceptions, self-report, observations)	21	31.34
Mixed Assessment	19	28.36

Note. Percentages may not sum to 100% due to rounding and the inclusion of studies in multiple categories (e.g., overlapping participant types, disciplines, or use cases).

generation (n = 20, 30%), supporting divergent thinking and exploratory learning. Feedback and reflection tasks (n = 19, 28%) leveraged ChatGPT for formative assessment, peer review, and metacognitive engagement, while writing support (n = 12, 18%) focused on drafting, summarizing, and revision. Instructional integration into flipped or blended classrooms (n = 8, 12%) positioned ChatGPT as a preparatory or co-instructional tool, and problem-solving tasks (n = 5, 7%) highlighted its use in analytical reasoning and computational thinking. In these contexts, ChatGPT functioned as more than a content generator—it was embedded as a pedagogical scaffold, dialogic partner, or ideation tool, prompting shifts in instructional sequencing, learner agency, and feedback mechanisms.

4.1.7. CCT assessment strategies

The included studies employed a wide range of methods to assess students' CCT. Across the full sample, 27 studies (40%) used direct assessments, such as standardized instruments (e.g., California Critical Thinking Skills Test, Torrance Tests of Creative Thinking, Alternative Use Test) and performance-based tasks (e.g., essays, digital product evaluations, argumentation coding) scored by experts or using rubrics. Indirect assessments were employed in 21 studies (31%), relying on participants' self-reported perceptions, reflective essays, Likert-scale surveys, thematic coding of qualitative data, or instructor observations. Mixed approaches were used in 19 studies (28%), combining quantitative scales with qualitative analyses such as thematic coding, reflective narratives, and text-based analyses. Notably, direct assessments were more prevalent in studies focusing on CrT, whereas indirect assessments dominated in studies focusing on CT. Studies addressing both constructs typically adopted mixed-methods designs, reflecting the multidimensional nature of CCT.

Taken together, these characteristics show that current evidence on ChatGPT's cognitive influence is shaped by a relatively narrow but intensively studied segment of higher education. The dominance of Asian institutions and STEM and teacher education programmes means that most findings arise from systems and disciplines where higher-order thinking is already foregrounded. At the same time, the prevalence of exploratory, mixed-methods designs and the sharp rise in publications since 2023 indicate a field still in an early mapping phase rather than one dominated by large-scale causal evaluations. Finally, the asymmetry in assessment strategies—where CrT is more often evaluated with direct performance tasks and CT with indirect self-report measures—suggests that apparent differences in reported robustness across the two domains partly reflect how they are operationalized and measured. This, in turn, may help explain why some studies report stronger evidence for CrT gains than for CT gains, independent of actual cognitive effects. These contextual and methodological regularities provide important background for interpreting the CT and CrT patterns reported in subsequent sections.

4.2. The impact of ChatGPT on students' CT across instructional contexts

Across the 56 studies examining CT, ChatGPT's cognitive influence emerged as a dynamic interplay of gains and risks, with outcomes heavily shaped by instructional design. Five principal themes captured how ChatGPT supported CT development, while five recurrent patterns highlighted its potential to hinder cognitive engagement. These effects spanned disciplines, task types, and educational levels and were consistently moderated by pedagogical framing, learner characteristics, and the instructional roles assigned to the AI (Table 5).

4.2.1. Affordances: ChatGPT as a catalyst for CT

A recurring finding (n = 27) emphasized ChatGPT's role in fostering metacognitive engagement. Structured activities such as guided prompting, comparative analysis, and reflective writing enabled students to monitor their thinking, exercise evaluative judgment, and develop critical skepticism. These strategies supported iterative self-evaluation, particularly when students examined differences between their outputs and those generated by ChatGPT (e.g., Strobl et al., 2024; Tseng & Lin, 2024; Zhan & Yan, 2025).

Another frequent affordance (n = 22) involved enhanced argumentative reasoning. In writing-intensive and debate-oriented tasks, ChatGPT facilitated the organization of arguments, integration of evidence, and refinement of claims. Studies employing argument mapping and rubric-guided evaluation reported significant improvements in the sophistication of student reasoning, with ChatGPT acting as a dialogic scaffold or counterargument generator (e.g., Chen et al., 2025; Darmawansah et al., 2025; Sykes, 2024).

In 19 studies, ChatGPT use prompted error detection and verification behaviors. Particularly in STEM and interdisciplinary domains, students engaged in fact-checking and triangulation, often using AI-generated content as a stimulus for skeptical inquiry. These behaviors were reinforced when students were explicitly taught to identify hallucinations or inconsistencies (e.g., Archila et al., 2024; Michalon & Camacho-Zuñiga, 2023; Šedlbauer et al., 2024).

Support for self-regulated learning (SRL) was also prominent (n = 17). When learners interacted with ChatGPT through structured prompts, revision cycles, or inquiry-based tasks, they demonstrated planning, monitoring, and strategic adjustment—behaviors aligned with

Table 5
ChatGPT's impact on critical thinking: Key themes.

Theme	Representative Contexts	Moderators/ Conditions	Illustrative Studies
Affordances			
Metacognitive Engagement (n = 27)	Reflective writing, peer comparison, revision	Guided prompts, structured tasks	Strobl et al., 2024; Tseng & Lin, 2024; Zhan & Yan, 2025
Argumentative Structuring (n = 22)	Debate, argument mapping, academic essays	Iterative use, rubric alignment	Sykes, 2024; Chen et al., 2025; Darmawansah et al., 2025
Verification and Error Detection (n = 19)	STEM assignments, interdisciplinary writing	AI literacy, critical stance	Michalon & Camacho-Zuñiga, 2023; Archila et al., 2024; Šedlbauer et al., 2024
Self-Regulated Learning (SRL) (n = 17)	Feedback cycles, inquiry tasks	Prompt engineering, reflection	Yang et al., 2025; Lee et al., 2024; Shen & Chen, 2025
Disciplinary Reasoning (n = 15)	Systems design, entrepreneurship, scientific critique	Task authenticity, disciplinary alignment	Kofahi & Husain, 2025; George-Reyes et al., 2024; Exintaris et al., 2023
Limitations			
Overreliance and cognitive laziness (n = 21)	Essay writing, academic tasks	Limited training, novice users	Sarwanti et al., 2024; Villarino, 2025; Azmi et al., 2024
Surface-Level Engagement (n = 18)	Summarization, content generation	Lack of scaffolding, minimal reflection	Dilekli & Boyraz, 2024; Monib et al., 2025; Al-Alami, 2024
Erosion of Argument Development (n = 14)	Academic essays, discussion tasks	Simplified tasks, low complexity	Fan et al., 2025; Ravšelj et al., 2025; Esmaeil et al., 2023
Metacognitive Offloading (n = 12)	Multi-stage writing, reflective activities	Absence of SRL prompts	Fan et al., 2025; Wang et al., 2024; Yang et al., 2025
Inadequacy for Higher-Order Reasoning (n = 10)	Advanced reasoning, Socratic tasks	Tool limitations, task demands	Urhan et al., 2024; Fakour & Imani, 2025; Lee et al., 2024

SRL models. These effects were most pronounced in contexts featuring explicit metacognitive scaffolds and guided AI use (e.g., Lee et al., 2024; Shen & Chen, 2025; Yang et al., 2025).

Finally, in 15 studies, ChatGPT enabled domain-specific reasoning, especially in scientific, technical, or entrepreneurial settings. Without explicit instruction in CT, students engaged in complex analysis when ChatGPT was integrated as a co-developer, critique target, or ideation partner within authentic disciplinary tasks (e.g., Exintaris et al., 2023; George-Reyes et al., 2024; Kofahi & Husain, 2025). These findings highlight how critical engagement can emerge organically through task demands when AI is embedded in discipline-grounded inquiry.

4.2.2. *Limitations: barriers and risks to CT development*

Despite these affordances, a parallel set of concerns was identified—particularly in unguided or minimally scaffolded contexts. The most commonly reported limitation (n = 21) was cognitive offloading through overreliance. Students described reduced motivation for independent analysis and a tendency to defer to ChatGPT, especially among novice users, non-native speakers, or those lacking academic writing or AI literacy (e.g., Azmi et al., 2024; Sarwanti et al., 2024; Villarino, 2025).

Closely related was the issue of superficial engagement (n = 18). In the absence of structured pedagogy, students frequently accepted AI outputs without critical appraisal, leading to shallow interaction and reduced epistemic vigilance—particularly in summarization, essay writing, or content-generation tasks (e.g., Al-Alami, 2024; Dilekli & Boyraz, 2024; Monib et al., 2025).

In 14 studies, ChatGPT use was associated with erosion of argumentative development. Although the tool eased task completion, it often replaced the cognitive struggle integral to constructing and refining ideas. As a result, student reasoning became procedural and reliant on pre-formulated AI suggestions (e.g., Fan et al., 2025; Ravšelj et al., 2025; Esmaeil et al., 2023).

Twelve studies documented metacognitive offloading, where students relied on ChatGPT for feedback and revision but failed to engage in deeper planning or evaluative reflection. Particularly in multi-stage writing tasks, AI use led to surface-level edits at the expense of structural coherence and rhetorical purpose (e.g., Fan et al., 2025; Wang et al., 2024; Yang et al., 2025).

Finally, a foundational limitation concerned ChatGPT's epistemic boundaries. Ten studies noted that the model struggled to support advanced rationality, such as maintaining logical consistency, challenging assumptions, or sustaining Socratic dialogue. These deficits reflected structural constraints of the model itself rather than misuse by students (e.g., Fakour & Imani, 2025; Lee et al., 2024; Urhan et al., 2024).

Patterns in the CT literature revealed a consistent relationship between instructional design, learner profile, and cognitive outcomes. Studies reporting gains in metacognitive regulation, argumentative reasoning, verification, and self-regulated learning typically embedded ChatGPT in structured tasks requiring students to compare, justify, or revise AI outputs, supported by prompts, rubrics, or reflective scaffolds. In contrast, cognitive offloading, superficial engagement, and weakened argumentation clustered in unstructured settings—particularly in summarization and essay writing—where ChatGPT was used as a convenience tool. These risks were most pronounced among novice writers, non-native speakers, and students with limited AI literacy, whereas advanced or trained users were more likely to demonstrate sustained evaluative engagement. Overall, CT outcomes were closely tied to how ChatGPT was positioned in the task sequence—whether as a dialogic resource or a content provider—and to the degree of epistemic and metacognitive support. These instructional contrasts were consistent across disciplinary and regional contexts.

4.3. *The impact of ChatGPT on students' CrT across instructional contexts*

Across the 44 studies examining CrT, ChatGPT's cognitive influence reflected a complex interplay of enhancement and inhibition. These effects were shaped by instructional design, task demands, and usage patterns, with five recurrent affordances and five recurring constraints emerging across disciplines including language education, STEM, social sciences, and the creative arts. Outcomes were consistently moderated by task structure, the role assigned to ChatGPT, and the presence or absence of instructional scaffolds (Table 6).

4.3.1. *Affordances: ChatGPT as a facilitator of CrT*

The most frequently reported affordance (n = 31) involved ChatGPT's support for idea generation, extension, and reframing. Students engaged the tool as a partner in brainstorming and ideation, particularly during the early stages of writing, project planning, or narrative development (e.g., George-Reyes et al., 2024; Sarwanti et al., 2024; Villarino, 2025). Across both qualitative reflections and performance-based assessments, ChatGPT was credited with helping students surface “unique insights” and explore alternative perspectives (e.g., Essel et al., 2024; Urban et al., 2024; Xu, 2025).

Another common affordance (n = 24) was its role in scaffolding and elaborating creative output. ChatGPT assisted students in structuring content, experimenting with tone, and enhancing stylistic expression across genres and disciplines. This was especially salient in academic and EFL writing contexts, where students used the tool to refine coherence, develop paragraph flow, and test stylistic variations

Table 6
ChatGPT's impact on creative thinking: Key themes.

Theme	Representative Contexts	Moderators/ Conditions	Illustrative Studies
Affordances			
Ideation and Divergent Thinking (n = 31)	Essay writing, project design, creative problem-solving	Self-directed use; early-phase ideation	Sarwanti et al., 2024; Villarino, 2025; George-Reyes et al., 2024; Urban et al., 2024; Xu, 2025
Structural and Expressive Scaffolding (n = 24)	EFL writing, academic essays, revision tasks	Language scaffolds; iterative drafting	Avsheniuk et al., 2024; Kurt & Kurt, 2024; Tseng & Lin, 2024; Werdiningsih et al., 2024
Dialogic Engagement and Perspective-Shifting (n = 18)	Argumentation, debate, inquiry tasks	Prompt design; reflection cycles	Lee et al., 2024; Darmawansah et al., 2025; Li et al., 2025; Cake, 2025
Affective and Motivational Activation (n = 16)	Storytelling, scenario design, reflective writing	Expressive task framing; narrative openness	George-Reyes et al., 2024; Gervacio, 2024; Jayasinghe, 2024; Alzubi et al., 2025; Kofahi & Husain, 2025
Instructionally Mediated Gains (n = 21)	Flipped classrooms, design-based courses	Scaffolded frameworks (e. g., ADDIE, i4C)	Li, 2023; Abdelmagid et al., 2025; Urban et al., 2024; Cake, 2025
Limitations			
Overreliance and Creative Passivity (n = 20)	Essay writing, general academic tasks	Low AI literacy; unstructured use	Awal, 2024; Azmi et al., 2024; Zhang et al., 2024; Mejri et al., 2024
Loss of Voice and Affective Authenticity (n = 15)	Reflective writing, narrative tasks	Identity-based writing; lack of style training	Monib et al., 2025; Al-Alami, 2024; Kurt & Kurt, 2024
Suppression of Iterative Exploration (n = 13)	Script writing, ideation tasks	Lack of prompting or feedback cycles	Cake, 2025; Boers et al., 2025; Azmi et al., 2024
Risk-Related Inhibition (n = 11)	Assignments requiring originality	Psychological and privacy concerns	El-Akhras et al., 2024; Selem et al., 2025; Mejri et al., 2024
Instructional Deficit (n = 14)	Writing, inquiry, general academic tasks	No guidance, reflection, or prompt training	Mejri et al., 2024; Al-Alami, 2024; Zhang et al., 2024

(Avsheniuk et al., 2024; Kurt & Kurt, 2024; Tseng & Lin, 2024; Werdiningsih et al., 2024).

In dialogic and reflective tasks, ChatGPT promoted cognitive flexibility and perspective-taking (n = 18). When students engaged in argument construction, role-based simulations, or iterative prompt design, the tool functioned as a co-designer or dialogic stimulus—encouraging exploration from multiple vantage points (e.g., Darmawansah et al., 2025; Lee et al., 2024; Li et al., 2025). These affordances were amplified in activities such as debate, script development, and inquiry-based problem solving, where students responded to AI feedback by revising or reimagining their ideas.

Affective and motivational benefits were also evident (n = 16). Students frequently described ChatGPT as reducing creative anxiety and enhancing their willingness to initiate complex or expressive tasks. Especially in storytelling, scenario design, and open-ended writing, ChatGPT was perceived as a "brainstorming buddy" that made creative engagement more approachable and less intimidating (Alzubi et al., 2025; George-Reyes et al., 2024; Gervacio, 2024; Jayasinghe, 2024; Kofahi & Husain, 2025).

Importantly, the presence of instructional mediation significantly amplified creative outcomes (n = 21). When integrated into structured learning environments—such as flipped classrooms, ADDIE-based design courses, or scaffolded creative modules—ChatGPT use was associated with statistically significant gains in originality, fluency, and elaboration (e.g., Abdelmagid et al., 2025; Cake, 2025; Li, 2023; Urban et al., 2024). These gains were consistently linked to pedagogical elements including prompt training, iterative feedback loops, and reflective scaffolds.

4.3.2. Limitations: barriers and constraints to CrT

Despite these benefits, five recurring limitations were observed—particularly in contexts lacking structured guidance. The most prevalent constraint (n = 20) involved creative passivity driven by overreliance. Many students reported a diminished inclination to explore original ideas, defaulting instead to AI-generated content. This substitution of cognitive effort with convenience was especially pronounced in essay writing and general academic tasks (e.g., Awal, 2024; Azmi et al., 2024; Mejri et al., 2024; Zhang et al., 2024).

A second limitation was the erosion of personal voice and emotional authenticity (n = 15). In tasks demanding narrative identity or expressive writing, students noted that ChatGPT's outputs lacked individual style and emotional nuance. Attempts to integrate AI-generated suggestions often diluted the originality and affective resonance of their work (Al-Alami, 2024; Kurt & Kurt, 2024; Monib et al., 2025).

In 13 studies, researchers observed suppression of exploratory and iterative thinking. Absent explicit prompts to revise or critique ChatGPT's suggestions, student engagement often remained at a superficial level. Behavioral data revealed repetitive use and uncritical adoption, with minimal evidence of conceptual recombination or idea improvement (Azmi et al., 2024; Boers et al., 2025; Cake, 2025).

Risk-related constraints also emerged (n = 11), with quantitative data indicating that perceived risks—such as privacy concerns, psychological discomfort, or fear of underperformance—were negatively correlated with CrT outcomes (El-Akhras, Abd El-Wahab, Saghier, & Selem, 2025; Selem et al., 2025; Mejri et al., 2024). These concerns discouraged open-ended exploration and reduced students' willingness to take intellectual risks.

Finally, the absence of instructional scaffolding (n = 14) was frequently linked to limited creative outcomes. In unguided use contexts, ChatGPT often functioned as a passive answer provider. Without reflective prompts or structured interaction, students lacked the strategic framing necessary to engage in deep or divergent thinking (Al-Alami, 2024; Mejri et al., 2024; Zhang et al., 2024).

Collectively, CrT gains were most consistently reported when ChatGPT was introduced early in creative tasks and framed as a provisional resource to be adapted, recombined, or contested. These conditions—often implemented through flipped classrooms, design-based learning, or scaffolded creative modules—enabled students to explore ideas iteratively, experiment with style, and expand the solution space, yielding documented improvements in fluency, elaboration, and originality. In contrast, creative passivity, loss of voice, and limited iteration clustered in tasks where ChatGPT was used late or without guidance, typically as a generic writing assistant. Risk-related concerns—such as academic integrity, privacy, or performance anxiety—further suppressed CrT engagement, even in well-designed contexts. Overall, ChatGPT functioned as a facilitator when positioned as a generative partner in open-ended, low-stakes work, and as a constraint when treated as a source of finished output in high-stakes or tightly specified tasks.

4.4. Co-occurrence of CCT in ChatGPT-Integrated learning

Building on the CT- and CrT-specific patterns outlined above, this section synthesizes findings from studies examining the co-occurrence of CT and CrT outcomes in ChatGPT-integrated higher education.

Specifically, it investigates whether ChatGPT supports both domains simultaneously, privileges one over the other, or impedes both—and how these outcomes are moderated by instructional conditions. The-matic analysis revealed three dominant trajectories of co-occurrence: (1) mutually reinforcing development of CT and CrT, (2) asymmetrical enhancement of CrT alongside stagnation or decline in CT, and (3) joint erosion of both cognitive domains. Table 7 summarizes these patterns along with associated pedagogical moderators and illustrative studies.

4.4.1. Synergistic enhancement of CT and CrT

A substantial subset of studies (n = 18) reported positive and mutually reinforcing co-occurrence of CT and CrT, particularly when ChatGPT was embedded within thoughtfully designed instructional contexts. Evidence for this synergy was drawn from standardized assessments, expert evaluations of student work, qualitative analyses of learning processes, and students' self-reported cognitive engagement. Pedagogical designs such as inquiry-based, project-based, and iterative learning—enriched by dialogic interaction and reflective scaffolding—enabled students to simultaneously engage in analytical reasoning and creative idea generation.

For instance, Essel et al. (2024) demonstrated that integrating ChatGPT into classroom activities produced significant improvements in CT (e.g., critical openness and reflective skepticism) and CrT (e.g., courage, flexibility, innovative search, and inquisitiveness), supported by both quantitative outcomes and student reflections. Similarly, Li et al. (2025) examined the InquiryGPT system, which combined scaffolded prompts, reflective questioning, and guided exploration. This structured approach enhanced CT through evaluative reasoning, fostered CrT through idea synthesis, and strengthened problem-solving across STEM-based inquiry activities.

4.4.2. Asymmetrical gains: CrT enhanced, CT diminished

A smaller but conceptually important group of studies (n = 8) documented asymmetrical co-occurrence patterns, wherein ChatGPT tended to facilitate CrT—especially in terms of idea generation and fluency—while simultaneously inhibiting CT development. In these cases, creative benefits were often accompanied by reduced depth of reasoning, overreliance on AI outputs, or disengagement from critical evaluation.

For example, Wang et al. (2024) found that while students valued ChatGPT's assistance with ideation and structural organization in writing tasks, they perceived a decline in CT, citing generic content, limited originality, and shallow reasoning. Similarly, Azmi et al. (2024) observed that ChatGPT supported exploratory learning but diminished critical engagement, as students often uncritically accepted responses ("just follow what the AI says"). Aldulaijan and Almalki (2025) reported that postgraduate students used ChatGPT to accelerate creative ideation and build confidence, but their engagement often lacked depth, raising concerns about superficial understanding and weakened CT.

Klimova and de Campos (2024) likewise found that while students appreciated ChatGPT's role in idea development, they expressed concerns about its potential to impede independent reasoning and problem-solving. Sarwanti et al. (2024) echoed this pattern: students noted increased productivity and support in writing tasks but reported decreased motivation for critical inquiry. Additional evidence was provided by Habib et al. (2024), who documented notable gains in divergent thinking alongside concerns about diminished metacognitive engagement; Esmaeil et al. (2023), who observed surface-level writing improvements coupled with reduced critical reflection; and Liu et al. (2024), who found that while GenAI tools improved writing mechanics, they provided limited scaffolding for CT development.

Across these studies, the primary sources of evidence were students' reflective narratives and self-reports, occasionally supported by analysis of academic outputs. Collectively, these findings suggest that although ChatGPT may stimulate creative processes, its unstructured use can coincide with erosion in CT—unless pedagogical strategies explicitly support balanced cognitive engagement.

4.4.3. Joint cognitive erosion: suppression of CT and CrT

A small but noteworthy group of studies (n = 4) reported negative co-occurrence patterns, wherein ChatGPT use was associated with simultaneous stagnation or decline in both CT and CrT. These outcomes were most common in settings where ChatGPT was used with minimal guidance or critical framing.

Ravšelj et al. (2025), based on a global survey of over 23,000 students, found that although ChatGPT was appreciated for its efficiency and summarization capabilities, students viewed it as largely ineffective for fostering higher-order thinking, citing its association with surface-level engagement and academic dishonesty. Al-Alami (2024) similarly reported that EFL learners valued ChatGPT's capacity to improve grammatical accuracy and save time, but noted no significant improvements in writing quality or critical-creative output. Students explicitly expressed concerns that reliance on AI might hinder long-term development of CT and CrT.

Zhang et al. (2024) further identified that students reporting AI dependency experienced increased intellectual passivity, reduced creativity, and diminished capacity for critical, independent thought. In these studies, evidence was predominantly derived from large-scale survey data and student self-reports, with some supplemented by analysis of student writing or performance artifacts. Collectively, these findings underscore that without instructional scaffolding, ChatGPT integration may inhibit both divergent and evaluative dimensions of student thinking.

Taken together, the findings offer a comprehensive map of both the empirical terrain and the cognitive dynamics associated with ChatGPT integration in higher education. The 67 included studies—spanning diverse research designs, disciplinary areas, and global regions—reflect a surge of scholarly interest in how generative AI reshapes students' thinking. Across this literature, ChatGPT's cognitive impact emerges as neither uniformly positive nor inherently negative, but contingent on the instructional contexts in which it is embedded. Positive outcomes—including gains in metacognitive regulation, argumentative reasoning, divergent thinking, and expressive fluency—were most consistently observed in pedagogical settings that emphasized

Table 7
Co-occurrence patterns in ChatGPT-Integrated learning.

Pattern Type	Thematic Description	n	Moderators/Conditions	Illustrative Studies
Synergistic Enhancement (CT↑, CrT↑)	ChatGPT facilitated simultaneous gains in CT and CrT by supporting inquiry, reflection, analysis, and ideation through structured learning models.	18	Inquiry-oriented tasks, scaffolded reflection, dialogic interaction, iterative refinement	Essel et al. (2024); Li et al. (2025); Chen et al. (2025); Darmawansah et al. (2025)
Asymmetrical Development (CrT↑, CT↓)	ChatGPT enabled fluency, ideation, and exploration, but lowered critical engagement due to shallow prompts, lack of evaluation, or passive tool use.	8	Unstructured implementation, emphasis on ideation over evaluation, minimal critical framing	Wang et al. (2024); Azmi et al. (2024); Aldulaijan and Almalky (2025); Klimova and de Campos (2024); Sarwanti et al. (2024); Habib et al. (2024); Esmaeil et al. (2023); Liu et al. (2024)
Joint Cognitive Erosion (CT↓, CrT↓)	CT and CrT outcomes both stagnated or declined, often in unguided or instrumentalist contexts with minimal cognitive challenge.	4	Passive/unscaffolded usage, absence of metacognitive prompts, task-completion focus	Ravšelj et al. (2025); Al-Alami (2024); Zhang et al. (2024); Monib et al. (2025)

scaffolded inquiry, reflective engagement, and dialogic interaction. Conversely, risks such as cognitive dependency, loss of narrative authenticity, and the joint erosion of higher-order thinking predominated in contexts where ChatGPT was used passively, without critical framing, structured prompts, or creativity-centered supports. Co-occurrence patterns further revealed that while the synergistic development of CT and CrT is attainable under well-designed, inquiry-oriented conditions, unstructured implementation frequently led to asymmetrical or diminished cognitive engagement.

5. Discussion

This review synthesized 67 empirical studies to examine how ChatGPT shapes students' CCT in higher education. Anchored in a dual-lens framework and informed by an integrated constellation of educational technology theories, the findings reveal that ChatGPT's cognitive effects are not intrinsic to the tool itself but arise from its interaction with instructional design, learner agency, and epistemic framing. Rather than serving as a uniformly beneficial or detrimental resource, ChatGPT functions as a cognitive mediator whose outcomes are contingent on how its affordances are mobilized within structured pedagogical ecologies. Given the predominance of short-term, non-experimental designs and methodological heterogeneity across the included studies, we interpret the evidence primarily as patterns of association and moderation rather than definitive causal mechanisms.

Across the reviewed studies, the most consistent divider was not the presence of ChatGPT but the instructional ecology surrounding it. When tasks and classroom norms required verification, justification, and iterative revision, students were more likely to treat AI outputs as provisional representations to interrogate—supporting co-occurring gains in critical evaluation and creative exploration. When such norms were weak, fluent outputs often lowered perceived task difficulty and encouraged premature closure, shifting regulation from monitoring and strategy use toward acceptance and cognitive offloading. Importantly, these tendencies reflect the broader activity system in which ChatGPT is embedded (e.g., task objects, feedback structures, and attribution/verification rules), rather than properties of the tool alone.

5.1. ChatGPT as a cognitive amplifier in scaffolded contexts

Across the reviewed studies, innovative instructional models redefined how generative AI is embedded in learning environments. Rather than treating ChatGPT as a static content generator, these models positioned it as a dynamic element within scaffolded, dialogic inquiry processes. This marks a shift toward what might be described as AI-mediated pedagogy—an evolving set of practices in which students engage with AI outputs to externalize reasoning, refine ideas, and test solutions across academic domains. These configurations signal pedagogical innovation in both task design and classroom interaction. In guided learning environments—characterized by structured prompts, reflective scaffolds, and dialogic interaction—ChatGPT consistently supported higher-order thinking. Students demonstrated gains in metacognitive regulation, argument construction, and idea generation. Drawing on distributed cognition (Hutchins, 1995), ChatGPT can be interpreted as an interactive cognitive artifact that enables iterative reasoning when learners engage in cycles of generation, critique, and refinement. Especially in writing and inquiry-based tasks, learners externalized their thinking into revisable intermediate representations that were evaluated, reorganized, and justified across iterations, supporting both convergent judgment (critical thinking) and divergent exploration (creative thinking).

These affordances also map onto self-regulated learning models (Zimmerman, 2002), where structured engagement with ChatGPT promoted planning, monitoring, and evaluative judgment. Students leveraged the tool not merely to complete tasks but to enhance cognitive control, echoing prior work on AI-mediated metacognition (Azevedo et al., 2010; Chen et al., 2025). When pedagogical design encouraged epistemic reflection, this interaction supported students' regulatory routines (e.g., planning, monitoring, and evaluative judgment). Notably, these benefits were most evident when students were prompted to monitor uncertainty and justify revisions rather than accept fluent outputs at face value.

5.2. Instructional mediation and the Co-occurrence of CT and CrT

Variation in outcomes—whether synergy, asymmetry, or erosion of CT and CrT—was shaped by the presence or absence of instructional scaffolds. This pattern can be interpreted through Cultural Historical Activity Theory (CHAT) (Engeström, 1987), which helps analyse how ChatGPT may shift activity systems by redistributing cognitive labour, reshaping rules for verification and attribution, and reframing task objects toward iterative justification rather than answer production. When embedded in epistemically rich tasks, the tool supported both divergent ideation and critical evaluation, enabling students to navigate between exploration and justification. These conditions facilitated adaptive expertise (Hatano & Inagaki, 1986), where students generated original ideas and refined them through analytic reasoning. When course norms and feedback structures kept responsibility for checking, attribution, and justification with students (rather than implicitly transferring it to the tool), CT–CrT synergy was more stable. Such designs reflect early-stage innovations in educational practice, including epistemically rich prompt engineering, multi-perspective AI dialogue, and reflexive use of ChatGPT as both ideation partner and critical foil—all of which reconfigure the traditional separation between creative and analytical thinking.

In contrast, under unstructured conditions, this synergy often unraveled. Students either defaulted to creative fluency without critical filtering or relied on cautious reasoning devoid of imaginative risk. This imbalance underscores the fragility of CT–CrT co-development and highlights the importance of "epistemic friction" (Medina, 2013)—the productive tension that sustains higher-order cognition. Absent such friction, AI-generated fluency can bypass the reflective struggle central to deep learning. These patterns also point to the role of AI literacy—particularly epistemic vigilance and verification routines—in sustaining critical evaluation alongside creative exploration.

5.3. Semantic fluency and the limits of unstructured use

ChatGPT's linguistic fluency emerged as both a strength and a constraint. While it enabled rapid ideation and stylistic experimentation, it also risked masking epistemic gaps. In several studies, students accepted coherent outputs uncritically, resulting in superficial engagement. These outcomes illustrate both sides of cognitive load theory (Sweller, 2011): while ChatGPT can offload extraneous demands and free capacity for synthesis, this benefit depends on scaffolds that manage intrinsic complexity and promote deliberate processing. In the absence of such scaffolds, reduced perceived effort may encourage premature closure and overreliance—an instance of cognitive offloading rather than strategic load management.

In creative domains, students reported improvements in elaboration and fluency, yet often noted a dilution of personal voice or emotional nuance. This tension—between expressive support and narrative flattening—mirrors concerns raised in AI-supported composition (Kurt & Kurt, 2024; Monib et al., 2025). The findings suggest that while ChatGPT can scaffold stylistic refinement, it may simultaneously constrain authentic creative agency when uncritically adopted. This further underscores that fluency alone does not produce higher-order thinking; it must be coupled with evaluative norms and regulatory routines that preserve authorial agency and epistemic responsibility.

5.4. ChatGPT as a boundary object and connective node

Across disciplinary and linguistic contexts, ChatGPT operated as a boundary object (Star & Griesemer, 1989)—a flexible, adaptable artifact that maintained coherence while enabling contextual reinterpretation. Students used it to navigate genre conventions, translate between rhetorical registers, and adapt responses across tasks. This interpretive flexibility supports cognitive adaptability and creative recombination, as seen in studies involving translanguaging and narrative cohesion (Sarwanti et al., 2024; Tseng & Lin, 2024). However, the same flexibility can also amplify surface-level synthesis when verification norms and critical evaluation are not explicitly required.

From a connectivist lens (Siemens, 2005), this boundary-spanning function can be understood in terms of pattern formation across distributed networks. ChatGPT served as a generative node within these networks, facilitating idea integration across disciplinary and cultural boundaries. Especially in collaborative or interdisciplinary settings, its use was associated with enhanced pattern recognition, analogical thinking, and cross-contextual synthesis—hallmarks of CrT development in networked learning environments. When combined with structured critique and source triangulation, these networked recombinations were more likely to support the development of creative and critically defensible outputs.

These evolving roles position ChatGPT not only as a boundary object but as a catalyst for instructional innovation. Studies described practices such as AI-supported storyboarding, genre transformation exercises, prompt-based innovation challenges, and cross-disciplinary synthesis tasks—pedagogical formats that expand beyond conventional writing or problem-solving assignments. While not always framed as formal innovations, these patterns reflect a broader shift in educational design: from outcome-driven instruction to process-oriented, generative learning anchored in AI-human collaboration.

6. Implications, limitations, and directions for future research

6.1. Practical, pedagogical, and theoretical implications

The findings of this review offer a set of actionable implications for educators, instructional designers, and curriculum developers seeking to harness the cognitive potential of ChatGPT in higher education. Central to these implications is the recognition that ChatGPT's influence on students' critical and creative thinking is not intrinsic but shaped by pedagogical design, epistemic framing, and learner engagement. Informed by the integrated conceptual framework, this section articulates a set of pedagogically grounded strategies for supporting meaningful learning with generative AI. These strategies represent emerging innovations in AI-mediated pedagogy and can be further strengthened through educational data analytics—such as tracking prompt–response cycles, revision behaviors, and reflection patterns. Moreover, the implications are grounded in key educational theories, ensuring strong theoretical alignment with AI-enhanced instructional design.

6.1.1. Embed structured cognitive scaffolding aligned with task goals

To activate higher-order thinking, ChatGPT should be integrated into tasks with intentional scaffolds that align with distributed cognition and metacognitive regulation. Educators should implement stepwise activities that guide students through prompt design, output evaluation, and iterative refinement. For example, a writing assignment might require students to: 1) generate an outline using ChatGPT, 2) critique its structure against rubric criteria, and 3) revise the content based on both AI and peer feedback. Such cycles externalize reasoning and promote evaluative depth by framing ChatGPT as a dialogic partner rather than a solution provider. These scaffolded task designs not only support deeper thinking but also generate traceable behavioral indicators (e.g., prompt quality, revision paths, iteration sequences), which can be analyzed as part of pedagogical data analytics systems to monitor cognitive engagement in real time.

6.1.2. Explicitly teach AI literacy to cultivate epistemic vigilance

Students' engagement with ChatGPT varies widely depending on how its role is epistemically framed. Without explicit guidance, many learners tend to adopt fluent AI-generated content uncritically, leading to risks of misinformation, bias, or cognitive offloading. To mitigate these risks, instructors should integrate AI literacy modules into coursework that emphasize prompt refinement, hallucination recognition, bias detection, and contextual interpretation. These instructional practices not only foster epistemic vigilance, but also produce analyzable interaction data—such as prompt histories, revision behaviors, and reflection responses—that can inform adaptive feedback, personalized learning analytics dashboards, and intelligent tutoring systems. This approach is grounded in metacognitive and self-regulated learning theories (Azevedo et al., 2010; Zimmerman, 2002), which emphasize learners' capacity to monitor and adjust their strategies when interacting with complex tools like generative AI.

6.1.3. Design tasks that co-activate CT and CrT through recursive inquiry

Rather than isolating CT and CrT into separate skill sets, instructors should develop assignments that foster their synergistic interplay, consistent with theories of adaptive expertise and Cultural Historical Activity Theory (CHAT). Effective examples include: 1) Open-ended case studies requiring students to brainstorm solutions (CrT), then justify and compare alternatives (CT); 2) Argumentative writing tasks where students must generate multiple perspectives via ChatGPT and critique each position; and 3) Project-based tasks that involve creative ideation (e.g., narratives, prototypes) and analytical reflection on design decisions. In these tasks, ChatGPT serves as both a boundary object and an ideation scaffold, supporting divergent generation while enabling convergent evaluation. These recursive designs can produce rich learner data—such as drafting timelines, edit trails, and evaluative comment patterns—providing input for cognitive process analytics and supporting real-time feedback loops in intelligent learning environments.

6.1.4. Implement reflection protocols that promote cognitive regulation

The review identifies that gains in metacognitive engagement often result from structured reflection around ChatGPT applications. Instructors should embed guided reflection prompts after each interaction, such as: 1) “What part of this AI-generated response was most useful or misleading?” 2) “How did this output shape your thinking—and did you accept it too quickly?” and 3) “What would you change in your next prompt to get a more rigorous result?” Such prompts extend beyond task completion to support self-regulated learning, reinforcing evaluative reasoning and promoting epistemic accountability. Analyzing these reflection responses using NLP-based sentiment and epistemic stance analysis can generate learner profiles that inform adaptive feedback and help educators detect blind spots in reasoning or signs of overreliance on AI tools.

6.1.5. Leverage ChatGPT as a connective node for interdisciplinary thinking

Findings suggest that ChatGPT facilitates connectivist learning when students are prompted to synthesize ideas across disciplines. Instructors can design cross-domain tasks—such as ethical debates in science courses or data narratives in humanities contexts—that ask students to draw on ChatGPT's broad knowledge base while critically examining disciplinary assumptions. This promotes both CrT (through conceptual recombination) and CT (through critique of source logic), encouraging integrative, pattern-based thinking across academic boundaries. Concept-mapping tasks and cross-disciplinary prompt chains using ChatGPT can also generate networked learning traces, which serve as a foundation for connectivist learning analytics—revealing how students bridge domains and reframe ideas over time.

6.1.6. Position feedback as a multi-source process beyond the AI

Finally, ChatGPT should be nested within feedback-rich learning ecosystems. While AI can assist with formative commentary, optimal cognitive growth occurs when its feedback is triangulated with peer review, instructor input, and self-assessment. To mitigate overreliance on AI-generated output, instructors should explicitly require students to justify revisions, document decision rationales, and verify claims using independent sources and human feedback. Instructors should design tasks with layered feedback checkpoints—e.g., initial AI draft, peer response, instructor annotation—so students iteratively calibrate and refine their ideas through multiple perspectives. These multi-source feedback loops can be logged and analyzed to track students' epistemic calibration, revision justifications, and responsiveness to critique, offering educators actionable behavioral analytics that complement qualitative assessment.

6.2. Limitations

This review is subject to several limitations that inform priorities for future research. First, the composition and scope of the evidence base introduce constraints on breadth and representativeness. The review was limited to peer-reviewed, English-language journal articles published between 2022 and April 2025, potentially excluding relevant studies from non-English contexts or those reporting null or negative findings. This may have narrowed the epistemic diversity of the dataset and skewed perceptions toward more visible or positive results. Additionally, publication trends in this emerging field—driven by heightened interest in AI and innovation—may contribute to selection bias, with studies demonstrating favorable outcomes more likely to appear in the literature. Many included studies were also conducted by early adopters operating in technologically advanced or pedagogically progressive settings, further limiting the generalizability of findings across broader educational contexts.

This scope decision also reflects a broader tension in systematic reviews of rapidly evolving fields: the imperative for comprehensiveness versus the need for stable, peer-reviewed evidence and feasible boundaries. While prioritizing journal publications supported methodological rigor and quality assurance, it excluded potentially relevant findings from high-quality conference proceedings, such as Learning Analytics and Knowledge (LAK), Artificial Intelligence in Education (AIED), and Learning at Scale (L@S). These venues often disseminate cutting-edge methods and pilot evidence ahead of journal publication cycles; their exclusion may therefore have limited coverage of the most recent methodological innovations and emergent findings. At the same time, conference papers vary in peer-review rigor, version control, and accessibility, which complicates quality appraisal and replication. By anchoring the synthesis in journal articles, we prioritized depth of analysis, reproducibility, and interpretive stability.

Second, the methodological and interpretive heterogeneity across studies poses challenges for synthesis and inference. Included studies varied widely in design, from qualitative case studies to exploratory quasi-experiments, and employed diverse definitions, assessments, and uses of ChatGPT. While this diversity enabled a rich thematic synthesis, it constrained cross-study comparability and limited the possibility of drawing causal conclusions. Most studies lacked control groups, standardized instruments for assessing CCT, or longitudinal tracking. The narrative synthesis approach, though appropriate for integrating heterogeneous data, is inherently interpretive and subject to researcher bias. Coding decisions were cross-validated and resolved through consensus, but subjective judgments may have influenced the construction and emphasis of thematic categories. Furthermore, although instructor mediation and learning-environment design appear to be central moderators, they were not the primary analytic focus of this review and were too inconsistently operationalized across studies to support fine-grained synthesis. Taken together, these features suggest that our synthesis should be understood as exploratory and hypothesis-generating rather than as providing causal estimates or statistically generalizable evidence of ChatGPT's effects on critical and creative thinking.

Third, the rapidly evolving nature of generative AI introduces temporal and technological limitations. The studies reviewed reflect interactions with different versions of ChatGPT, each with varying capabilities in terms of fluency, reliability, and user guidance. As the tool continues to develop—becoming more multimodal, customizable, and embedded within educational platforms—some findings may become outdated or require reinterpretation. More broadly, the pace of research on ChatGPT in education means that any review risks obsolescence: between our search date (April 19, 2025) and the present, additional studies have likely appeared that are not captured here. This temporal instability poses challenges for generalizing results and highlights the need for ongoing empirical work that tracks affordance shifts over time.

6.3. Future directions

These limitations point to several critical priorities for future research. First, there is an urgent need for longitudinal and experimental studies to assess the sustained impact of ChatGPT use on students' critical dispositions, creative agency, and metacognitive regulation. Most current studies adopt cross-sectional or short-term designs, which limits understanding of how cognitive habits evolve with continued AI interaction. Equally pressing is the absence of theory-driven research. Despite widespread reference to “thinking skills,” few studies explicitly drew on established frameworks—such as dual-process theory, epistemic cognition, or self-regulated learning—to operationalize or interpret CCT outcomes. Future investigations should be anchored in robust theoretical models that can clarify the mechanisms by which generative AI shapes higher-order cognition.

Second, methodological innovation is needed both in primary studies and in evidence synthesis to keep pace with a rapidly evolving technological environment. Generative AI tools undergo frequent updates, often with limited transparency or backward compatibility, creating challenges for replication, standardization, and meaningful comparison across studies. Researchers will require agile and adaptive designs—such as mixed-methods approaches, learning analytics, and embedded practitioner–researcher partnerships—to capture tool use in context and trace real-time learner–tool interactions. Future syntheses should also consider integrating findings from high-quality, peer-reviewed conference proceedings (e.g., LAK, AIED, L@S) to capture methodological innovations and pilot studies that often emerge ahead of journal publication cycles. Living systematic reviews that incorporate continuous updating protocols may offer a more sustainable approach to evidence synthesis and better track shifts in pedagogical design, student outcomes, and disciplinary variation while maintaining methodological rigor (Elliott et al., 2017). Comparative meta-syntheses contrasting journal and conference findings may further clarify how publication venue shapes the framing, rigor, and dissemination speed of educational AI research.

Beyond these foundational concerns, several underexplored areas merit deeper investigation. Comparative studies across academic disciplines are needed to understand how domain-specific norms, epistemologies, and assessment practices mediate students' engagement with ChatGPT. Research should also examine how learners' epistemic beliefs, AI literacy, and metacognitive strategies influence their ability to engage reflectively with AI-generated content. These cognitive dispositions may shape not only learning outcomes but also the quality of students' reasoning and the depth of their inquiry.

Finally, the sociocultural, ethical, and identity-related dimensions of thinking in AI-mediated environments require further exploration. Questions surrounding authorship, originality, and academic integrity are central to how students navigate critical and creative tasks in the presence of generative AI. In parallel, global disparities in language access, infrastructure, and cultural norms of inquiry demand context-sensitive research that attends to equity, inclusivity, and local epistemologies.

Taken together, these priorities call for a new wave of research that is theoretically grounded, methodologically agile, and contextually responsive. Advancing this agenda will be essential to ensuring that the integration of generative AI into higher education promotes not only cognitive development, but also epistemic responsibility and educational equity.

CRediT authorship contribution statement

Changyue Li: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hang Cui:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Linda Serra Hagedorn:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

Consent to participate

Not applicable.

Ethics approval

This article is a systematic review of previously published studies and does not involve any new research with human participants or animals conducted by the authors. Accordingly, this review did not require institutional ethical approval. All included studies were peer-reviewed publications available in the public domain.

Consent for publication

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Materials and code availability

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2026.100571.

Data availability

All data analyzed in this review were extracted from publicly available, peer-reviewed publications. The synthesized data are fully reported within the manuscript and its tables/figures. Additional details about the data extraction and coding procedures are available from the corresponding author upon reasonable request.

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